

ENX MONEY — MASTER PENDING WORK ROADMAP

MOBILE OTP VERIFICATION & KYC COMPLIANCE IMPLEMENTATION MATRIX

Complete Technical Audit, DLT Regulations, DigiLocker KYC, Cloud Persistence & Go-Live Plan

Organization:	Enterprenex Solutions Pvt. Ltd.	Official Domain:	https://enxmoney.enterprenex.solutions
Project Name:	ENX Money (Digital Khata, GST & Billing)	Target Audience:	Leadership, Tech Leads, DevOps & QA
Document Version:	v4.7.0 (October 5, 2026)	Overall Status:	Core Ready (Final Production Steps Active)

1. EXECUTIVE SUMMARY

The ENX Money ecosystem has completed its frontend and web acceleration milestones: the high-converting OkCredit-style landing page is live with uninterrupted live video demonstration, social links are connected, Flutter Web load time is accelerated from 2–4 minutes down to 2–4 seconds, and all DPDP Act 2023 legal policies are active. This master report outlines the exact remaining technical steps required to transition ENX Money into commercial production, specifically detailing Mobile OTP verification, KYC compliance (DigiLocker, Setu, PAN & Bank Penny Drop), and cloud infrastructure.

COMPLETED & LIVE

92%

Web app, landing, legal suite, release APK/AAB.

MOBILE OTP VERIFY

85% Built

Engine active; needs Indian TRAI DLT registration.

KYC VERIFICATION

80% Built

DigiLocker/Setu sandbox ready; needs live keys.

REMAINING GO-LIVE

5 Tasks

Play Store, Cloud RDS, Webhooks, Backups, QA.

2. COMPLETED & VERIFIED PRODUCTION MODULES

Component	Status	Benchmark / Metric	Location / Route
Fintech Landing Page	100% Live	Modern OkCredit Style	https://enxmoney.enterprenex.solutions
Live Video Demo	100% Live	Continuous 16s Video Loop	Hero Section (Clean, No badges)
Official Social Media	100% Live	Direct LinkedIn & Instagram	Footer Links (Verified IDs)
Flutter Web Acceleration	100% Live	49ms /app, 1.6MB JS (w/o 7 EMP)	/app (CanvasKit Local + Gzip)
DPDP 2023 Legal Suite	100% Live	9 Refund, 13 Terms, 14 Privacy	/privacy-policy, /terms, /refunds
Google Play Account Deletion	100% Live	In-App & Public Web Form	/delete-account, DELETE /api/account
Production Android Binaries	100% Live	Signed APK (71MB), AAB (50MB)	ENX-Money.apk & ENX-Money.aab
Razorpay Payment Gateway	100% Live & Processing	Live Payments Verified & Active	rzp_live_ThjlhbHvQ4iaQV (Live)

3. MOBILE PHONE OTP VERIFICATION — ARCHITECTURE & DLT GUIDE

ENX Money supports both Email and Mobile Phone Number OTP verification. The backend codebase features an enterprise SMS service (server/src/services/sms.service.js) equipped with E.164 international phone number normalization, multi-provider routing (Fast2SMS, AWS SNS, Twilio), and cryptographic OTP generation with brute-force protection.

A. Current Architectural State:

- Codebase Support: AuthService.sendOtp and AuthService.verifyOtp accept phone as an E.164 string.
- Universal Test Bypass: OTP code 123456 is active for test review, Play Store approval, and developer testing.
- Rate Limiting: 60-second resend cooldown and 5-minute expiration prevent spam and brute-force attacks.
- Provider Routing: Server detects FAST2SMS_API_KEY, AWS_ACCESS_KEY_ID, or TWILIO_ACCOUNT_SID automatically.

B. Why Some Real Mobile SMS Fail Without DLT Registration in India (TRAI Mandate):

In India, the Telecom Regulatory Authority of India (TRAI) strictly requires all commercial and transactional SMS to pass through Distributed Ledger Technology (DLT) gateways. Messages with unregistered templates or header IDs are immediately dropped by telecom operators (Jio, Airtel, Vi, BSNL).

C. Exact Step-by-Step Action Plan to Enable 100% Real Mobile OTPs:

Step & Phase	Responsible Party	Action Required	Deliverable / Output
1. DLT Portal Registration	Enterprenex Solutions	Register on Jio / Airtel DLT Portal	Enterprise Entity ID & Certificate
2. Header (Sender ID)	Enterprenex Solutions	Register 6-character Header (e.g. ENXMMNY)	Approved Transactional Header
3. Content Template Approval	Enterprenex Solutions	Submit: "Your ENX Money code is {#var#}. Valid for 5	Approved Template ID
4. Fast2SMS / Twilio Config	DevOps / Lead	Add DLT Template ID to dashboard & recharge	Active SMS delivery routing
5. Render Environment Update	DevOps	Set FAST2SMS_API_KEY in Render Cloud Dashboard	Zero-failure real SMS dispatch

D. Instant Zero-DLT Alternative: Meta WhatsApp OTP Dispatch

Because ENX Money already has the Meta WhatsApp Cloud API integrated (server/src/services/whatsappStatementService.js), mobile OTPs can also be dispatched directly to the user's WhatsApp without requiring TRAI DLT registration. WhatsApp OTPs deliver in <1.5 seconds, cost ~1/3 of Indian SMS, and boast 99.8% open rates.

4. KYC VERIFICATION ENGINE — DIGILOCKER, SETU, PAN & BANK PENNY DROP

The ENX Money KYC Engine (server/src/services/kycProvider.service.js) provides automated Government of India identity verification to protect dukandaars and ensure compliance with RBI & DPDP Act 2023 financial guidelines. The engine supports a 3-pillar verification pipeline: PAN, Aadhaar via DigiLocker, and Bank Account Verification via Penny Drop.

A. The 3-Pillar KYC Verification Architecture:

- 1. NSDL / Income Tax PAN Verification:**
Validates business/individual 10-digit PAN against the official Income Tax database. Confirms active status and matches registered merchant name.
- 2. Aadhaar Verification via DigiLocker (MeriPehchaan):**
Connects to Government of India DigiLocker OAuth2 portal to fetch verified digital Aadhaar XML without storing sensitive biometric or unencrypted Aadhaar data.
- 3. Bank Account Verification (BAV Penny Drop):**
Transfers ₹1 via IMPS to the merchant's bank account to verify bank account existence, IFSC validity, and confirm the bank account title matches the PAN card.

B. Why KYC is Currently in "Sandbox / Pending" State:

The KYC backend currently runs in Setu Sandbox Mode (SETU_SANDBOX_GATEWAY) because live production API credentials from Setu or DigiLocker have not yet been generated by the organization and inserted into Render environment variables. In sandbox mode, test credentials pass verification, but real merchants cannot complete live KYC until production API credentials are configured.

C. Step-by-Step Action Plan to Make KYC 100% Live in Production:

KYC Requirement	Provider / Portal	Required Credentials / Action	Production Status
1. Setu KYC Production Account	Setu (Bridge/Pine Labs)	Sign agreement & obtain SETU_CLIENT_ID &	Pending Org Agreement
2. DigiLocker Redirect Callback	Render Cloud Server	Whitelabel https://enxmon- eventerprenex-solutions/	URL Ready for Setup
3. PAN Verification Activation	Setu / NSDL Gateway	Configure PAN_API_KEY in Render Cloud	Sandbox Active, Needs Key
4. Bank Penny Drop Activation	Setu / Razorpay Fund Acct	Deposit ₹500 float balance for ₹1 verification drops	Ready for Live Float
5. Merchant Tier Automation	ENX Money Backend	Tier 0 (₹10k/mo) -> Tier 1 (₹1L/mo) -> Tier 2 (Unlimited)	Built in UserModel

5. MASTER 7-NUMBER WORK ASSIGNMENT MATRIX

Task 1: Mobile Phone OTP & SMS Gateway (DLT / WhatsApp)

[Backend Engineer & Business Admin | HIGH]

- Register Entity on Jio/Airtel DLT portal; approve Header ID (e.g. ENXMNY).
- Approve OTP Template: "Your ENX Money code is {#var#}. Valid for 5 mins."
- Recharge Fast2SMS wallet & configure FAST2SMS_API_KEY on Render.
- Enable Meta WhatsApp OTP as zero-DLT instant delivery backup.

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Task 2: KYC Engine Part 1: PAN Card & Aadhaar via DigiLocker

[Backend Lead & Compliance Officer | HIGH]

- Complete Setu KYC enterprise account onboarding & sign production agreement.
- Configure SETU_CLIENT_ID, SETU_CLIENT_SECRET & SETU_PRODUCT_INSTANCE_ID on Render.
- Connect live NSDL PAN API to verify 10-digit PAN and match merchant name.
- Whitelist callback URL for Aadhaar OAuth: <https://enxmone.enterprenex.solutions/api/kyc/digilocker/callback>.

Task 3: KYC Engine Part 2: Bank Account Verification (₹1 BAV Penny Drop)

[Backend Engineer & Finance Ops | HIGH]

- Activate ₹1 IMPS bank penny drop to verify account holder name matching PAN.
- Deposit ₹500 float balance in Setu / Razorpay Fund Account for automated penny drops.
- Automate merchant Tier upgrade upon successful bank verification (Tier 1 limit: ₹1 Lakh/mo).

Task 4: Google Play Console Release & App Publication

[Android Developer & Product Manager | HIGH]

- Upload ENX-Money.aab (58.4 MB) to Play Console Internal/Production track.
- Upload 512x512 app icon, 1024x500 banner, and 4-6 app screenshots.
- Fill out Data Safety Form using [server/public/data-safety.html](https://play.google.com/about/data-safety/audit-faq) audit.
- Provide Google Play QA test account credentials (code 123456).

Task 5: Production Cloud Database Setup (AWS RDS / Managed MySQL)

[DevOps & Database Administrator | HIGH]

- Launch AWS RDS MySQL 8.0 instance in ap-south-1 (Mumbai).
- Run complete database schema migration from database/schema.sql.
- Update DB_HOST, DB_USER, DB_PASSWORD, DB_NAME on Render.
- Verify GET /health reports database: connected.

Task 6: Meta WhatsApp Permanent System User Token & Statement Engine

[Backend Integrations Engineer | MEDIUM]

- Generate Permanent System User Token in Meta Business Suite.
- Update WHATSAPP_ACCESS_TOKEN on Render to prevent 24h expiration.
- Test automated customer WhatsApp statement PDF dispatch.

Task 7: Automated Daily S3 Database Backups & Physical Multi-Device QA

[DevOps & QA Lead | HIGH]

- Set up daily midnight mysqldump cron to encrypted AWS S3 bucket (30-day retention).
- Install ENX-Money.apk on Android 11, 12, 13, 14 physical devices (Samsung, Redmi, OnePlus).
- Pilot onboard 5-10 local retail kirana dukandaars for field validation.

6. PRODUCTION GO-LIVE TIMELINE & SIGN-OFF MATRIX

Phase / Milestone	Target Window	Key Deliverable	Dependency
Phase 1: Google Play Store	Days 1 - 2	AAB Upload, Store Assets, Form Submission	Store Graphics (512x512, 1024x500)
Phase 2: Production DB	Days 2 - 3	AWS RDS MySQL Connected & Migrated	AWS Cloud Credentials / DB URL
Phase 3: DLT / WhatsApp OTP	Days 3 - 5	100% Real Phone OTP Delivery	TRAI DLT Template / Meta Token
Phase 4: Live KYC Activation	Days 4 - 6	Setu / DigiLocker Live Credentials Active	Setu Production API Keys
Phase 5: Webhooks & Settlement	Day 5	Automated Razorpay Webhook, Callbacks	Razorpay Dashboard Access
Phase 6: Commercial Launch	Day 7	Public Play Store Live & Pilot Onboarding	Google Play App Review Approval

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Project Lead / Director Sign-off: _____
Date: _____

